

Windows 11 readiness report — Halden device fleet

Prepared for: Managing Director and Finance Director **Prepared by:** Muhammad Ibrahim Akmal, IT ·
Date: 2026-10-02 **Supporting data:** data/synthetic-fleet.csv and data/synthetic-fleet-summary.md **Companion documents:** p04-exec-brief.md , p04-endpoint-standard.md

⚠ This fleet is SYNTHETIC

Halden is a fictional company and this report is a home-lab portfolio artefact. The device fleet described below is an **invented inventory of 150 machines**, generated deterministically to demonstrate a real readiness method. It is **not** a survey of real hardware, and no real device was measured. Every count in this report is a plain count of that synthetic file, produced by data/gen_synthetic_fleet.py , and is labelled as a synthetic-fleet figure throughout.

The *Rules* and the *method* are real: they are the actual Windows 11 requirements and the actual classification logic that scripts/07-Get-Win11Readiness.ps1 applies to real machines.

1. Executive summary

The synthetic fleet contains **150 devices**. Under the Windows 11 requirements:

Status	Devices	Share of fleet
ready — capable and already on Windows 11	78	52%
upgrade — capable hardware still on Windows 10	41	27%
replace — a hardware requirement is not met	31	21%

(Synthetic-fleet figures. Counts derived directly from data/synthetic-fleet.csv ; shares are simple arithmetic on those counts.)

The practical reading: roughly half the fleet needs nothing, roughly a quarter can be moved to Windows 11 with an in-place upgrade, and roughly a fifth cannot run Windows 11 at all and needs replacing. The Windows 10 population as a whole — devices needing an upgrade or a replacement — is ****72 of 150 devices (48%)****. Those devices currently receive no security updates from Microsoft.

Recommendation: Option A. Upgrade the 41 capable devices in place, and replace the 31 incapable devices to a schedule. It has the lowest ongoing risk and the fewest moving parts. Option B (Extended Security Updates as a bridge) is a fallback if the capital spend cannot be released this year, and it must be understood as a delay, not a solution. Option C is not a plan — it is accepting an unpatched estate.

2. Why this matters (business context)

- Windows 10 reached end of support on **14 October 2025**. Devices that remain on it receive no security fixes.
- Industry data shows about ****21%** of small-business Windows devices were still on Windows 10 in

mid-2026** (Lansweeper, reported by Computer Weekly). Halden's synthetic 48% is worse than that average, which is exactly why the question is being asked now rather than next year.

- Microsoft's commercial **Extended Security Updates** run for at most three years and must be paid for each year, and the annual price rises. ESU buys time; it does not remove the need to move.
- **Ransomware is the main risk for a business this size**, and it starts on endpoints. An unsupported device is the easiest place for it to start.
- Insurance and customer due-diligence questionnaires increasingly ask about the operating system lifecycle. "We are working on it" is a weaker answer than a dated plan.

3. Method

1. The Windows 11 requirements were written down once (`configs/win11-readiness-rules.yaml`): TPM 2.0, Secure Boot, UEFI firmware mode, a supported CPU (Intel 8th generation / AMD Zen+ or newer), at least 4 GB RAM and at least 64 GB system disk.
 1. The same rules were implemented once in code (`data/fleet_rules.py`), used by the generator, the analysis script and the unit tests, so the three cannot disagree.
 1. The synthetic fleet was generated deterministically (`data/gen_synthetic_fleet.py --count 150 --seed 42`), so anyone can reproduce the exact file and the exact counts.
 1. Devices were classified: any unmet hardware requirement means `replace` ; all met and still on Windows 10 means `upgrade` ; all met and already on Windows 11 means `ready` .
 1. The live inventory script (`scripts/07-Get-Win11Readiness.ps1`) applies the same logic to real lab machines, so the method is proven on hardware and only the fleet scale is synthetic.

4. Fleet status in detail (synthetic fleet)

4.1 By status

Status	Devices	Share	What it means for the business
<code>ready</code>	78	52%	No action. These devices are already compliant with the standard.
<code>upgrade</code>	41	27%	An in-place upgrade on a schedule; a short per-device disruption.
<code>replace</code>	31	21%	These cannot run Windows 11 and must be replaced to keep receiving security updates.

4.2 By department

Department	ready	upgrade	replace	Total
Finance	11	5	4	20
HR	5	1	3	9
IT	10	3	1	14
Management	10	5	3	18
Operations (incl. warehouse)	24	19	10	53
Sales	18	8	8	34
Shared / kiosk	0	0	2	2
Total	78	41	31	150

(Synthetic-fleet figures. The department split is part of the generated data, adjusted so that shared kiosks and warehouse terminals belong to no single team.)

4.3 Why devices are classified `replace` (synthetic fleet)

Hardware requirement not met	Devices affected
CPU older than Intel 8th generation / AMD Zen+	31
TPM 2.0 missing or disabled	10
Legacy BIOS (no UEFI firmware mode)	8
Secure Boot not enabled	8

A device can fail more than one requirement, so these counts add up to more than the 31 `replace` devices — which is the useful detail for planning, because a machine failing only the CPU check and one failing the TPM check need the same answer (replace), but the cause tells you how old the estate is.

4.4 Where the risk is concentrated

- **Finance (4 of 20 devices) and Management (3 of 18)** include a higher proportion of older hardware.

These are the devices most likely to hold sensitive data, so they should be replaced first.

- **Operations/warehouse (10 of 53)** carries the largest number of incapable devices, including shared

terminals and thin clients. These are the least personal and easiest to swap in a batch.

- **Shared kiosks (2)** are both incapable and are shared machines logging into shared accounts, so they

are the quickest, cheapest replacements.

5. Options for management

All three options cover the same 72 devices that currently receive no security updates (41 to upgrade, 31 to replace).

Cost note: the labour figures are planning estimates from the project's upgrade runbook, and the device and ESU figures are **placeholders** until a current vendor quote and the current Microsoft ESU price are attached. The report must state the quote date before it is used for a budget decision.

Option	What it involves	Indicative cost	Risk	Ongoing position
A. Upgrade and replace (recommended)	Upgrade the 41 capable devices in place this quarter; replace the 31 incapable devices over the next two quarters, Finance and Management first	Labour for 72 devices + 31 × device cost	Lowest. Ends the unsupported estate on a dated plan	A fully supported fleet within two quarters
B. Upgrade, then ESU as a bridge	Upgrade the 41 capable devices now; buy Extended Security Updates for the 31 incapable devices for one year and replace them next financial year	Labour for 41 devices + ESU year 1 + 31 × device cost later	Medium. ESU price rises each year and covers only three years; it delays, it does not remove, the spend	Still ending with a replacement programme, but a year later and at a higher total cost
C. Do nothing	Continue as today	No new spend now	High. Unpatched operating systems, uncontrolled data on unencrypted portable devices, and a question the business cannot answer in a due-diligence questionnaire	No date on which the estate becomes supported

5.1 Why Option A is recommended

- It is the only option that ends with the problem solved rather than deferred.
- Replacing the oldest devices first removes the machines least able to run modern security controls

several have no TPM 2.0 at all, which means no BitLocker and no Credential Guard either, so they weaken the endpoint standard as well as the patch position.

- The work is predictable: an upgrade per device is a short, repeatable task with a runbook, and replacements can be batched by department.

5.2 If Option B is chosen

Be explicit with the business about the trade: ESU is a **bridge**, not a plan. It is a defensible choice if capital cannot be released this year, provided the replacement programme is approved at the same time with a date, so the bridge has something on the other side.

6. Recommended plan and timeline

Phase	Weeks	Work	Owner
1	1-2	Approve Option A and the replacement budget; confirm vendor quotes and the ESU price for reference	MD + FD
2	1-2	Replace the 4 Finance and 3 Management incapable devices first	IT + Finance
3	3-6	In-place upgrade the 41 capable devices in department batches, using the upgrade runbook	IT
4	3-6	Replace the remaining 24 incapable devices (Operations/warehouse batch first, then Sales)	IT
5	7	Confirm the whole fleet passes the daily endpoint check; close the programme with a short report	IT

7. Risks and dependencies

Risk	Mitigation
A critical business application does not work on Windows 11	Test the specific applications a user runs during the upgrade pilot, not the full installed list; the upgrade runbook has a rollback window
Replacement hardware lead times slip	Order the Finance and Management devices first, since they carry the most sensitive data
Users are disrupted during the upgrade	Batches by department, agreed dates, upgrades done around work, and a rollback available for ten days
ESU price or eligibility is different from the placeholder	Confirm the current Microsoft price and eligibility before any Option B decision is taken
The readiness data is mistaken for a real survey	Every page of this report states that the fleet is synthetic; the live inventory script exists to produce real data on real machines

8. What we are asking for

#	Decision	Owner	Date required
1	Approve Option A and the replacement schedule	Managing Director + Finance Director	This meeting
2	Approve the replacement budget for 31 devices (31 × the quoted device cost)	Finance Director	This meeting
3	Agree the department order for replacement (Finance, Management, then Operations, then Sales)	Department heads	This week

Approval (unsigned until a review actually happens):

Managing Director: Date:

Finance Director: Date:

Lab note: Halden Distribution Ltd. is fictional and this report is a home-lab portfolio artefact. The fleet counts and percentages labelled "synthetic fleet" come from an invented dataset produced by a deterministic generator; they are reproducible by re-running the generator, and they are not a measurement of real hardware. All other content — the requirements, the method and the runbooks — reflects work that is real and executable.